BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 189 · Search and sort: comparisons and Big O

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What property does binary search need that linear search does not?

Q2. What time complexity is typical of merge sort?

Q3. For n=5, how many times does a single for-loop from 0 to n-1 run?

Q4. A nested loop runs 4 times outside and 4 times inside. How many inner operations occur?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

time complexity — How an algorithm's work grows with input size.

O(n) — Work proportional to input count.

O(log n) — Work proportional to the number of halvings.

O(n²) — Quadratic growth, often nested loops.

trade-off — A benefit balanced against another cost.

## New learning

Count what grows — Big O describes how work increases with input size, not the exact milliseconds on one computer.

Compare search — Linear search can take n checks; binary search on sorted input roughly halves the remaining candidates each step.

Compare sorting — A simple nested comparison loop can be O(n²), whereas merge sort is O(n log n).

## Worked example

For 16 values, compare maximum comparisons of linear and binary search.

1. Predict first — Assume one successful or unsuccessful lookup in sorted data.

2. Model solution — Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.

# We Do · Practise with guidance

Repeat the comparison for 64 ordered values.

Predict first

What happens when you double 32 to 64?

Model solution

Linear worst case = 64 comparisons; binary worst case at most 7. Binary grows much more slowly.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Name the complexity of a single loop visiting n values.

### Question 2

An array has 128 ordered items. Compare linear and binary worst-case checks.

### Question 3

Is binary search always better than linear search? Give one counterexample.

## Exit Ticket · Answer before checking

O(log n) means execution time is literally log n seconds.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Sorted input.

Q2. O(n log n).

Q3. 5 times.

Q4. 16 operations.

### You Do pages · Independent questions

Q1. O(n).

Q2. Linear up to 128 checks; binary up to 8 checks with a standard midpoint search.

Q3. No. An unsorted tiny array can be searched linearly without sorting. Binary search first needs sorted data, and sorting/preparation may cost more than one scan.

### Exit Ticket · Model answer

False. It describes the growth of operations, not a time in seconds.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**